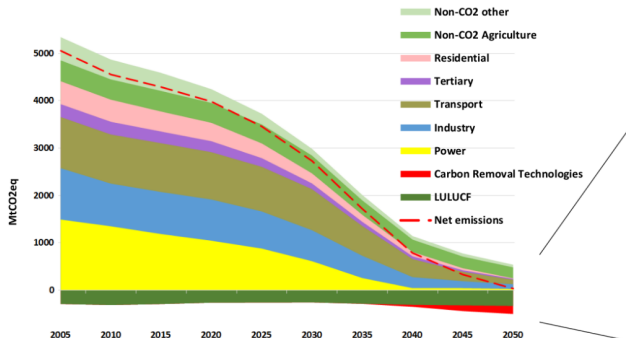


To reach net-zero emissions, all sectors need to decarbonise!

Paris-compliant 1.5° C scenarios from European Commission - **net-zero**



Wind and solar dominate the expandable potentials for low-carbon energy provision.

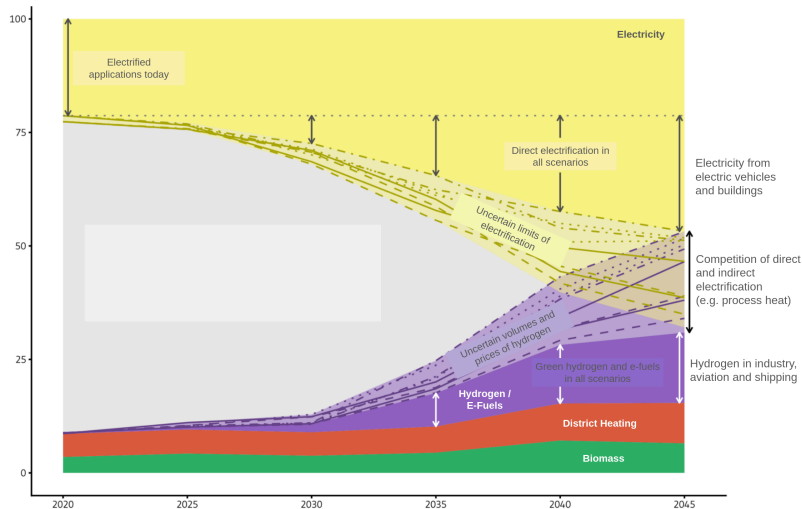
So **electrification** is essential to decarbonise sectors such as transport and heating.

Source: European Commission (2018),

https://ec.europa.eu/clima/sites/clima/files/docs/pages/com_2018_733_en.pdf

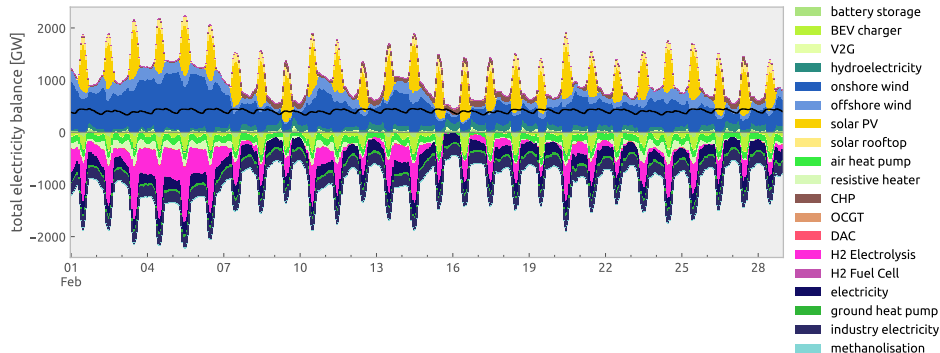
Many scenarios show increase in electrification across sectors...

Final energy shares [%]



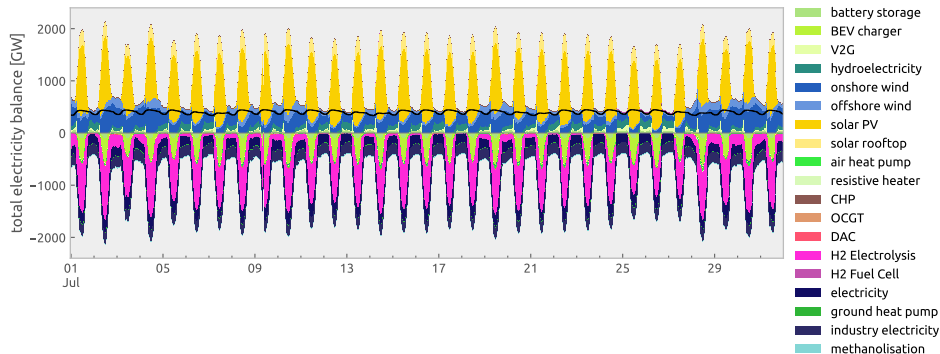
Source: Ariadne (2021), <https://ariadneprojekt.de/>

Electrification in many sectors affects operation of power system...



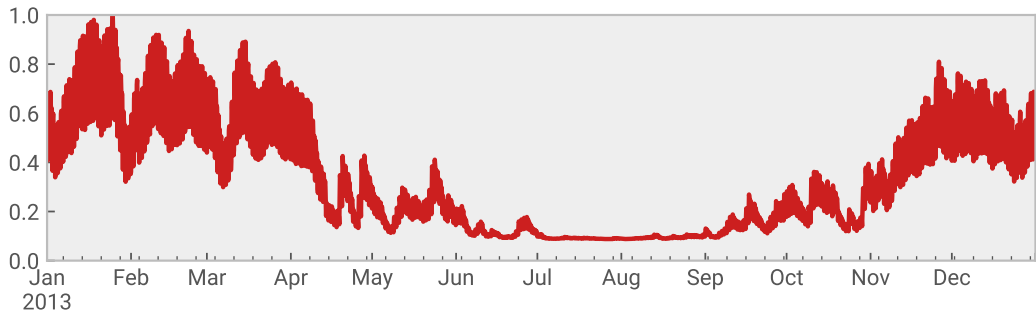
Sector-coupling adds a lot of electricity demand, requiring **much more wind and solar** buildout than if electricity sector alone were to be decarbonised.

...but also offers new opportunities for balancing renewables!



But sector-coupling also offers many ways to manage the variable renewable feed-in cost-effectively, through **demand-side management** (e.g. smart charging of electric vehicles) and **conversion to heat and chemical fuels** which are easier and cheaper to store

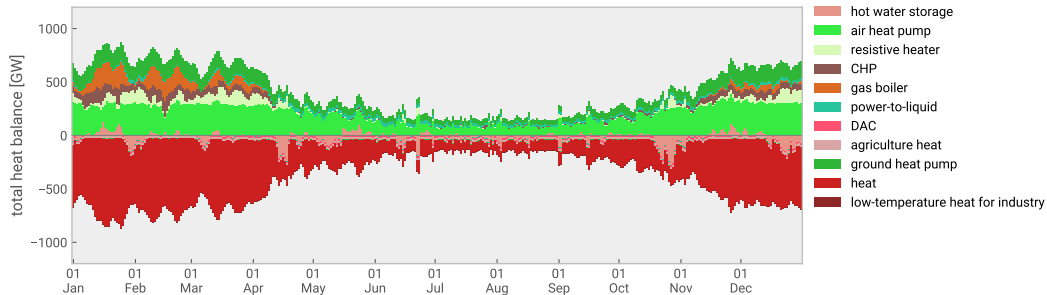
Time series of heat demand show strong seasonal variation.



Compared to electricity demand, heating demand is strongly **seasonally peaked**.

In summer, most heat demand is for **hot water** provision – in winter for **space heating**.

Example daily heat balance for net-zero emissions in Europe



There are difficult periods in winter with **low wind and solar generation**, **high space heating demand**, and **low air temperatures**, which are bad for air-sourced heat pumps!

In this case **gas boilers** and **CHP plants** jump in as backup.

We also have some waste heat from e-fuel production in district heating.

How to calculate heating demand?

Heating degree day (HDD) quantifies the energy demand for heating a building. It is derived from the **outside air temperature** (T) and put in relation to a given **temperature threshold** (e.g. 15°C). The heating requirements are considered to be directly proportional to the number of HDD at that location.

$$\text{heat demand}(T) \propto \max\{0, 15^{\circ}\text{C} - T\}$$

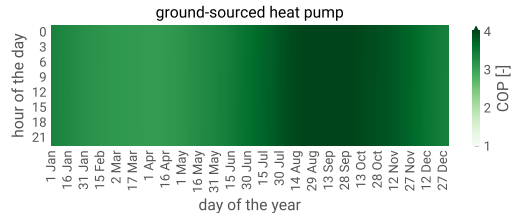
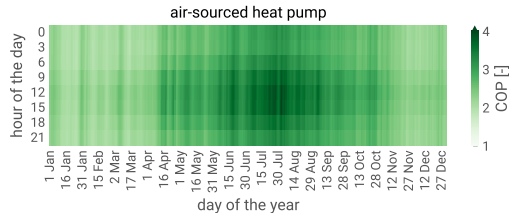
These daily profiles can be supplemented with typical hourly profiles for a weekday/weekend for downscaling. The baseline is typically a historical annual energy demand per region.

Heat pumps

Heat pumps use electricity to move heat from ambient air or groundwater (source) into the homes (sink).

The **coefficient of performance (COP)** is temperature-dependent and defined as:

$$COP = \frac{\text{thermal energy moved from source to sink}}{\text{input work (electricity)}} \propto \frac{1}{T_{\text{sink}} - T_{\text{source}}}$$



Heat pump performance is determined from measurements.

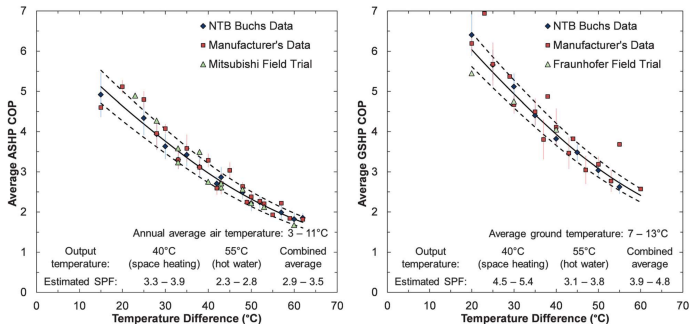


Fig. 9 Average heating coefficient of performance for air and ground source heat pumps (left and right, respectively) based on data taken from industrial surveys and field trials.^{31,40–42} The inset tables show the expected performance for UK conditions.

For instance, for an air-sourced heat pump an approximated function could look like this

$$COP(\Delta T) = 6.81 + 0.121\Delta T + 0.000630\Delta T^2$$

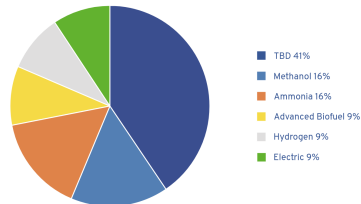
where ΔT is the difference between the air (source) and water storage (sink) temperature.

Transport subsectors

Land transport

- battery electric vehicles
- fuel cell electric vehicles ?
- internal combustion engines ??

Shipping



Aviation

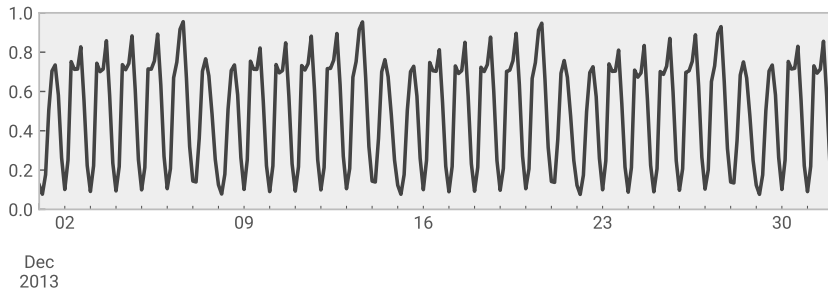
- electricity (short-haul to medium-haul)
- hydrogen (medium-haul)
- synthetic kerosene (long-haul)

Source:

<https://www.globalmaritimeforum.org/content/2022/11/The-2022-Annual-Progress-Report-on-Green-Shipping-Corridors.pdf>

To model demand-side management with electric vehicles,...

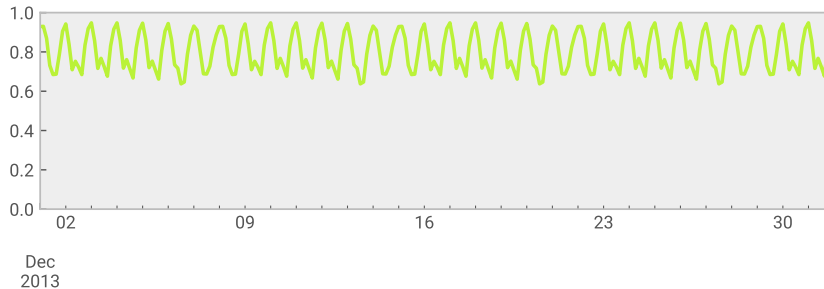
...need to know the typical mobility behaviour translating to energy consumption ...



(this profile is strongly peaked)

To model demand-side management with electric vehicles,...

...need to know the typical availability of electric vehicles for flexible operation ...

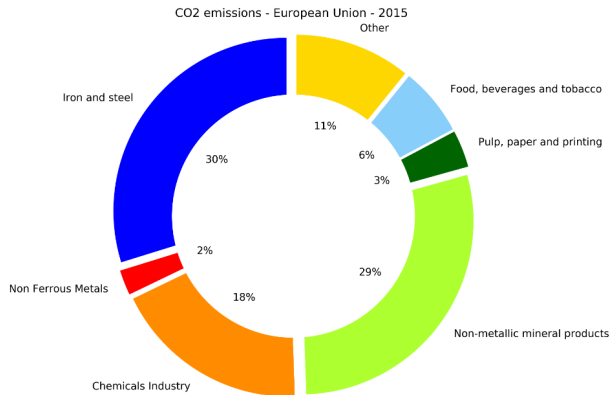


(this profile shows normalised availability for charging at home)

(need also technical parameters like charging speed and battery capacity)

(need also social parameters like when battery should be fully charged)

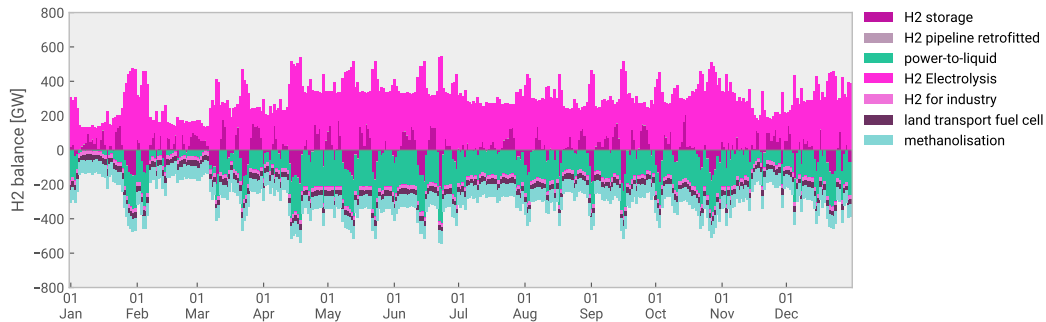
Which industry sector is responsible for most emissions?



- Non-ferrous metals: mainly aluminium, but also copper, lead etc.
- Non-metallic minerals: mainly cement, ceramics and glass
- Emissions come from combustion of fossil fuels for heat, as well as **process emissions** from chemical reactions

Industry: not only energy but also non-energy demands (feedstocks) are relevant.

H₂ – intermediate product between electricity and synfuels.

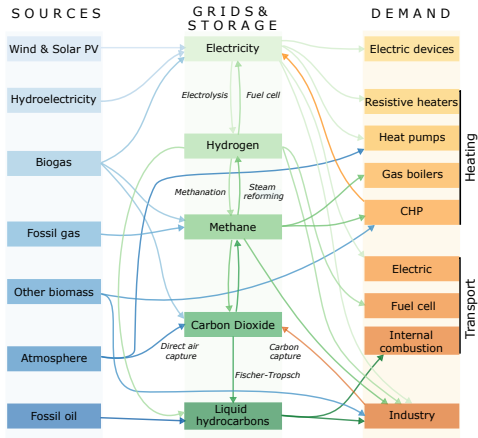


Relatively **few end-uses** of hydrogen (e.g. industry).

Most of the hydrogen is produced to produce **hydrogen derivatives**, e.g. high-value chemicals, kerosene, methanol, methane etc.

Many interactions between sectors that need to be modelled...

... each with sector-specific **energy demand patterns** and **flexibility options**.



- Electricity sector
- Transport sector
- Heating sector
- Industry sector
- Industrial feedstocks and biomass
- Electricity, hydrogen and gas networks
- Carbon management strategy

But modelling of components and flows is **not** fundamentally different from electricity system modelling!